

Developing a Perspectivist Historical Databank with Large Language Models

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Abstract

In this paper, we present Seshat-perspective, the first historical databank synthetically annotated with a perspectivist approach by means of Large Language Models. We propose a framework that utilizes Time-Resolved Variable prompting and three models, Llama 3.1 (US), DeepSeek R1 (China), and Mistral 3 (EU), to compare the historical knowledge they encode. The resulting dataset, which expands Seshat to 1023 polities fully annotated, is validated with transformer embeddings and correlation analysis, by comparing the model-generated descriptions and scores against 640 human-coded polities of the original Seshat databank, across key societal dimensions including warfare, religion, and social structure. Seshat-perspective and the code to generate new annotation layers are freely available, and can help researchers to spot cross-model discrepancies in historical knowledge.

Keywords

Cultural Analytics, Corpora, Cliodynamics, NLP for the Humanities

1. Introduction and Background

Recent scientific literature provided evidence that Large language Models (LLMs) encode the historical view of the culture that produced them, in particular for dimensions such as the type of elite in power in past societies [1]. Hence, exposure to AI-generated content from foreign models can potentially yield historical cultural hacking, the implicit distortion of specific historical information, with potential influence on collective memory [2]. Unfortunately, the geopolitical model origin alone cannot be used as a proxy for LLMs' worldview, as differences between models could come from many factors: technical (i.e.: model size, distillation, alignment, post-training filtering, content moderation) and cultural (e.g.: training languages, western bias, censorship). Although technical factors are easy to trace in LLMs, their cultural dimension is not. A recent survey on more than 90 works [3], categorized the probing techniques used to evaluate cultural representation of LLMs into two approaches: White-box probing (i.e.: analysis of the embedding space) and Black-box probing (e.g.: analysis of data generated by models). While White-box probing is possible only with open models, Black-box probing is always possible, and a big challenge is to find replicable frameworks to generate data and analyze it.

This paper describes a Black-box probing method to generate, evaluate, and analyze historical data with a synthetic perspectivist approach. In Natural Language Processing, perspectivism refers to preserving and modeling disagreement among human annotators [4], especially when disagreement is meaningful due to annotators' different backgrounds or interpretive positions [5]. In Natural Language Processing, perspectivism has been successfully applied to subjective tasks such as hate speech detection [6], obtaining useful insights on the classification decision process. The novelty of synthetic perspectivism is that the modeled disagreement is among LLMs of different origins and characteristics, rather than human annotators. The proposed method also evaluates LLMs' outputs against human-curated historical data, available in the Seshat Databank. This should not be treated as an agreement with historical truth, but rather as an agreement between LLMs and human experts.

The Seshat Project [7] played a fundamental role in the production and dissemination of quantitative

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historical data. The Seshat Databank [8] has been manually compiled by experts with a great deal of effort in time and cost. Between 2011 and 2025 the Seshat team identified more than 840 polities from 10 macro-regions, equally distributed over the world, in a time window spanning roughly from 10000 BC to 1900 CE. However, only about 640 polities are at least partially annotated. The data produced contain information about social complexity [9], axial age [10] and historical agricultural productivity [11], among others. The databank is freely available via the website¹ or APIs², and is released under a Creative Commons license³. Seshat contains textual descriptions of polities, and many different fine-grained variables that report the presence or absence of a cultural trait in a polity at a specific point in time. For example the presence/absence of copper, fortifications, firearms, written literature, coins and many others. These variables are designed to be treated computationally as boolean values with One-Hot Encoding (OHE), a widely used approach for expressing the presence or absence of categorical features into numerical ones represented as booleans. The Seshat team made a great effort to produce a codebook with replicable instructions for experts to compile the datasets [12]. Despite this, subjectivity cannot be eliminated when coding historical data with human experts, and critics argued that turning historical knowledge into boolean features is not a wholly objective enterprise [13].

While the creators of historical databanks are interested in using LLMs to extract information from historical sources in a faster way, users asking questions about historical events get responses from the internal knowledge of LLMs. Crucially, no dataset containing the historical knowledge encoded into LLMs exists. Moreover, LLMs have proven to be a cheap and fast solution for data annotation [14]. The first attempts to use LLMs for historical labeling designed prompts to ask single questions in a four-choice format, and LLMs showed a balanced accuracy ranging from 33.6% (Llama-3.1-small, 8 billion parameters) to 46% (GPT-4-Turbo, 8 expert models with 222 billion parameters each), outperforming random guessing (25%) but falling short of expert comprehension [15]. Artificial Intelligence evolves really fast, and recent research demonstrated that models above 400 billion parameters, such as Llama3.1-large, have human-level performance in historical data interpretation and annotation [16]. However, the annotation task for LLMs needs to be designed differently from humans. Seshat encodes boolean variables that create very sparse data, which is computationally inefficient to generate and to use [17]. Time Resolved Variable (TRV) prompting [18, 19] offers a way to make synthetic data generation more efficient for LLMs by requiring models to output concepts aggregated on temporal scales, suitable for historical annotation. In practice, TRV prompting works like a codebook, but provides an easier and faster way to define and test annotation instructions.

This paper introduces a new dataset, called Seshat-Perspective, that exploits TRV prompting and multiple annotations of LLMs developed in different countries, to extend Seshat towards a perspectivist approach, exploiting the internal historical knowledge of LLMs. The dataset is freely available online at https://github.com/facells/fabio-celli-publications/raw/refs/heads/main/docs/seshat_perspective.zip, and released under Creative Commons Attribution Non-Commercial (CC BY-NC SA) licensing. The data generation procedure is described in the Method Section; the Data Records Section describes the corpus, and the Technical Validation Section reports the evaluation procedure, which is quantitative and compares to the existing Seshat human-annotated data. Finally, the Conclusion Section provides a general summary of the findings and considerations about the usability of the presented dataset.

2. Method

The adopted method is depicted in Figure 1. It aims to generate three types of data about historical polities: (i) numerical variables about advancements in key dimensions such as infrastructure, information management, etc.; (ii) textual descriptions about the polity and (iii) textual references to literature that support the textual descriptions. The procedure starts from the polities that are present in the original Seshat databank (839) and their metadata, such as start time, end time and world-region cluster. These

¹<https://seshat-db.com/>

²<https://seshat-db.com/api/>

³<https://creativecommons.org/licenses/by-nc-sa/4.0/deed.en>

are clusters of modern States (i.e: "Israel/Palestine, Jordan, Lebanon, Syria") that provide a geographic reference to the polities of the past.

The procedure includes three steps: (i) *polities augmentation* consists in asking an LLM to suggest polities that are not present in the original data, in order to fill all temporal gaps within a world region cluster.

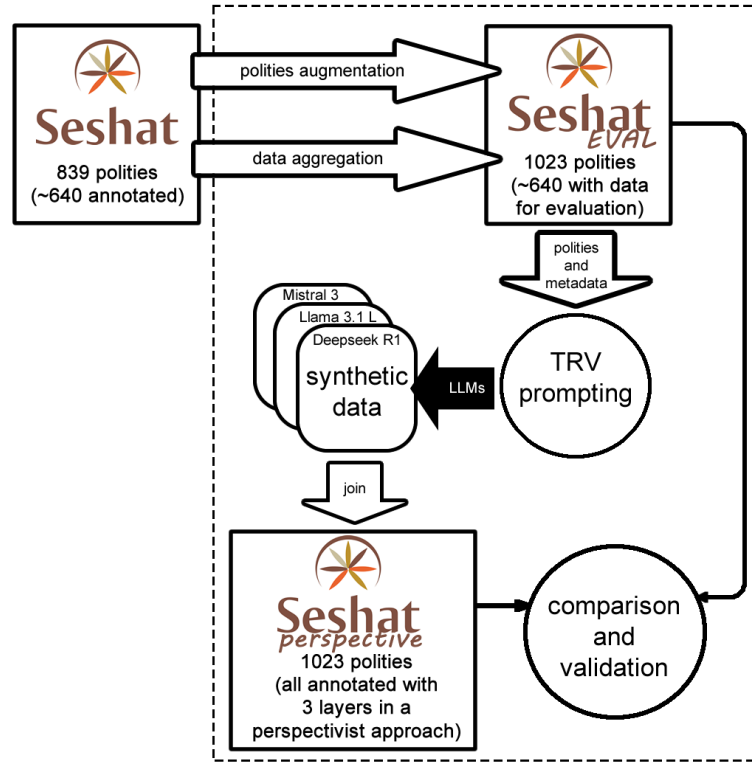


Figure 1: Schema of the method adopted for the generation and validation of Seshat-Perspective.

(ii) *Data aggregation*: the human-compiled data available in different original Seshat replication datasets have been aggregated in order to obtain a single test set. (iii) *Synthetic data generation*: polity names and some metadata such as polity location and time span have been used as input for TRV prompting in order to generate synthetic numerical variables, descriptions and references with Mistral 3 medium, Llama 3.1 large and Deepseek R1. Finally, these three layers of synthetic data are joined to create Seshat-Perspective, and the quality of the data produced has been evaluated against the human-coded variables, descriptions and references present in Seshat-eval. The annotation with LLMs is done on the whole augmented corpus, also filling in the incomplete parts. Each step is described in detail below.

Polities Augmentation The data augmentation strategy has been employed at the level of world region cluster and augmented the number of polities from 839 to 1023. A list of polities for each region cluster was provided to Gemini 2.0 flash through the prompt in Figure 2, receiving as output new polities existed in the regional cluster but not covered in the list.

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Act as an expert historian and generate a jsonl with information about the polities existed in
<regioncluster> between 10000 BC and 2000 CE, that are not in this list: <listofSeshatpolities>. Consider
States, Kingdoms, cultures, chiefdoms and type sites. Output the information with polity name, start
date, end date and region cluster as in the following example: { "polityname": "Canaan", "startyear":
"-2000", "endyear": "-1175", "regioncluster": "Israel/Palestine, Jordan, Lebanon, Syria"}
  
```

Figure 2: Prompt for polity augmentation.

A manual check was done to spot and discard duplicates and hallucinations. The distribution of polities per contemporary country before and after the augmentation is reported in Figure 3.

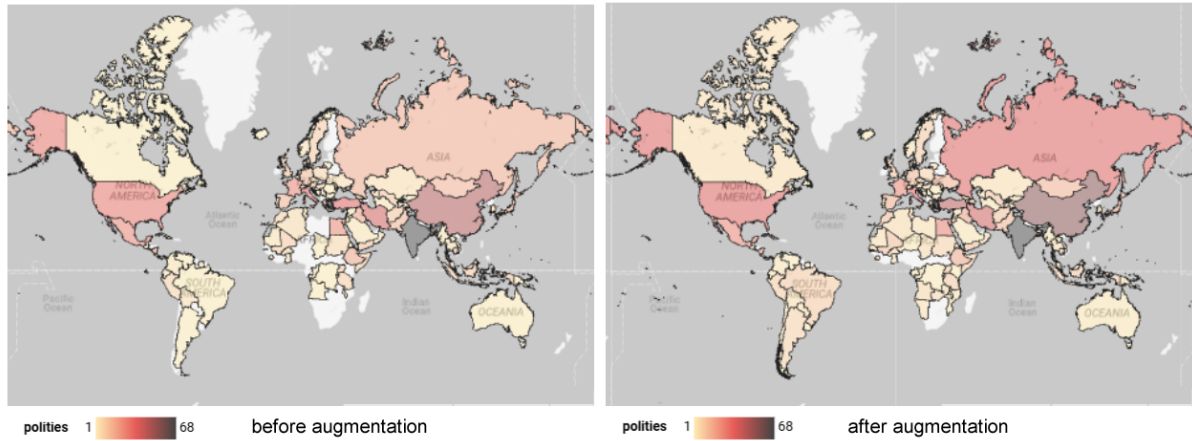


Figure 3: Distribution of sampled polities by contemporary country before and after polity augmentation. The process added polities in many countries such as Australia, Russia, China and Brazil. Among new countries covered there are Namibia, Niger, Mozambique, Libya and Chile.

Data aggregation and Seshat-eval Seshat-eval is a test set resulted from the aggregation of data contained in Seshat. Using polity identifiers and names, different tables have been joined in a single one with the following numerical variables:

- Polity Population (PolPop): the estimated logarithm of the total population of the polity.
- Polity Territory (PolTerr): the estimated logarithm of the total territory of the polity in square kilometers.
- Capital Population (CapPop): the estimated logarithm of the population of the polity's capital city.
- Government Complexity (gov): the one-dimensional, normalized Principal Component of the variables related to government specialization. This includes boolean values such as the presence of professions and institutions, measuring the complexity and sophistication of the polity's governmental organization.
- Infrastructure Complexity (infrast): normalized one-dimensional principal component measuring the extent and complexity of the polity's infrastructure (e.g., roads, irrigation, public works).
- Text/Record Keeping Complexity (texts): a one-dimensional normalized Principal Component of boolean values about the presence of texts (fiction, scripts, lists, etc.) measuring the complexity of record-keeping and bureaucratic use of texts.
- Monetary Complexity (money): a standardized index measuring the presence and complexity of a monetary system (e.g., use of fiat, commodity, or standardized currency).
- Agricultural Productivity (agriprod): estimated agricultural production in kilograms per hectare.
- Religious Levels (religlev): estimated number of command levels in religious hierarchies.
- Administrative Levels (adminlev): estimated number of command levels in administrative hierarchies.
- Military Levels (militlev): estimated number of command levels in military hierarchies.
- Settlement Levels (setlev): estimated number of settlement tiers (cities, towns, villages, hamlets etc.).
- Crises Consequence Index (crisis): first Principal Component from the boolean values contained in the Consequence of Crisis dataset (i.e.: decline, collapse, epidemic, downward elite mobility, etc.).

- The SPC1 Social Complexity Index (complexity): the normalized Principal Component derived from a set of variables related to social complexity. This summarizes data about polity size (population, territory and scale), government (professions, institutions) and hierarchy (command levels and settlement tiers), and is often used as a single, synthetic measure of a polity’s overall complexity score [20].

Beside these, Seshat-eval contains metadata about polities (id, polity code, polity name, start year, end year, region cluster, region, state code) and textual data for the evaluation, for instance human-coded polity descriptions and human-coded references to scientific literature.

Data generation and TRV Prompt The numerical values are generated through Time-Resolved Variables (TRVs), which encode sequences of categories onto numerical scales.

This scaling is determined either chronologically—based on the order of historical appearance, or by any rational scheme that orders concepts along a gradient. For example, a chronologically ordered TRV for military technology might assign 0.1 for stone/wood, 0.2 for copper, 0.3 for bronze, and 0.4 for iron, and so forth. Conversely, a TRV for social structures could use a rational scale, assigning 0.1 for democratic/participatory societies, -0.1 for hierarchical/autocratic societies, and 0 for mixed structures. Interpolation is incorporated using decimal numbers between these primary scale steps, which allows for the representation of transitional concepts between one level and the next. TRVs can be treated as equidistributed sequences or – when including information about the conceptual distance between steps – as low-discrepancy sequences. Research suggests that using TRV prompting as input to LLMs is comparable to a feature compression technique, making it computationally efficient for data generation and highly suitable for visualizations [18]. In order to have meaningful comparisons, the variables to include in Seshat-Perspective have been selected on the basis of the variables that are available for comparison in Seshat-eval, described in the previous paragraph. The prompt for data generation includes also the request to generate textual descriptions and literature to support these descriptions. The final prompt for generating synthetic historical data with the three LLMs is reported in Figure 4.

Three LLMs produced in different countries have been employed in order to obtain a diversified generation of information with the same prompt. Among LLMs developed in the US, Llama 3.1-large (405 billion parameters) [21] has been selected for its proven ability to annotate historical descriptions with Structural-Demographic phase labels at human level [16]. Among Chinese models, Deepseek R1 has been selected for its size (670 billion parameters) and reasoning capabilities [22]. Among the few European models, Mistral 3 medium (195 billion parameters) was selected because it showed excellent abilities in an expertise-demanding domain such as the medical one [23].

3. Data Records

The Seshat-Perspective corpus is freely available online⁴ as a package containing both Seshat-Perspective and Seshat-eval. Both datasets are structured as JSONL object (a single line contains a JSON file), where each line represents a historical polity. An example of Seshat-eval is reported in Figure 5.

The structure of Seshat-perspective data records can be broken down into three main sections, as reported in Figure 6: metadata, TRVs and textual data.

Metadata include identifiers, temporal and geographic location of polities, such as "id", "name", "new name", "long name", "start year", "end year", "duration", "century", "regioncluster" and "statecode". The TRVs refer to quantitative data generated by three LLMs, Deepseek R1 (dr1), Llama 3.1-large (l3l), Mistral 3 medium (m3m). Finally, the qualitative textual data includes "notes" and "references" in three versions generated by the three LLMs. There are three identifiers in Seshat-Perspective: "id" is a numerical identifier compatible with the API version of Seshat. Numbers below 0 are added during the augmentation phase and can be filtered out if required; "name" is the polity code compatible with early Seshat replication datasets; "new name" is the latest polity identifier in Seshat; and "long name" is the

⁴https://storage.googleapis.com/openrepo/seshat-perspective/seshat_perspective.zip

Act as an expert historian and generate a jsonl with information about the *<polityname>* existed in *<region>* between *<startyear>* and *<endyear>*. Consider that negative numbers are years BC. The json must have the following values:

"pop": polity population: 0=around 100 people, 0.1=between 100 and 1000 people, 0.2=between 1000 and 5000 people, 0.3=between 5 and 10 thousand people, 0.4=between 10 and 50 thousand people, 0.5=between 50 and 100 thousand people, 0.6=between 100 thousand and 1 million people, 0.7=between 1 and 100 million people, 0.8=between 100 and 500 million people, 0.9=between 500 million and 1 billion people, 1.0=population above 1 billion

"terr": polity territory class: 0.1=less than 100 km2, 0.2=between 100 and 500 km2, 0.3=between 500 and 1000 km2, 0.4=between 1 and 10 thousand km2, 0.5=between 10 and 50 thousand km2, 0.6=between 50 and 100 thousand km2, 0.7=between 100 and 500 thousand km2, 0.8=between 500 thousand and 1 million km2, 0.9=between 1 and 10 million km2, 1.0=more than 10 million km2

"war": level of technology and warfare: 0.1=stone/distant attacks, 0.2=copper/armies/fortifications, 0.3=bronze/chariots, 0.4=iron/siege engines, 0.5=steel/incendiaries, 0.6=gunpowder/logistics, 0.7=nuclear/cyberwarfare

"rel": religion type: 0.1=cult of the dead/spirits, 0.2=cult of ancestors/home/totem, 0.3=cult of fertility (father sky/mother earth), 0.4=polytheism, 0.5=monotheism, 0.6=phylosophy (buddhism, confucianism, taoism etc.), 0.7=ideology (capitalism, communism etc..)

"soc": social structure type: 0.1=egalitarian/democratic, 0.0=mixed, -0.1=stratified/autocratic, -0.2=totalitarian

"pow": elite power type: 0.0=infrastructure control, 0.1=war/force, 0.2=information/opinion management, 0.3=economy/production/wealth

"infrastr": level of infrastructures: 0.1=routes/quarries, 0.2=common/special buildings, 0.3=production systems/irrigation, 0.4=urbanisation/markets, 0.5=portual good supply systems, 0.6=railways/telecommunications, 0.7=space supply system

"infomedia": level of mediated information: 0.0=presence of oral tradition/gossip, 0.1=presence of symbolism/pictograms, 0.2=presence of calendars/writing/laws, 0.3=presence of religious/philosophical texts, 0.4=presence of literature/fiction, 0.5=presence of newspapers/press, 0.6=presence of computers/AI

"economy": level of economy: 0.0=subsistence/exogamy, 0.1=barter with goods/livestock, 0.2=exchanges with precious metals by weight, 0.3=coins, 0.4=paper currency, 0.5=stock market

"agri": type of agriculture: 0=spontaneous/no agriculture, 0.1=swidden/slash&burn, 0.2=fallow, 0.3=two fields/crop rotation, 0.4=use nitrogen-fixers/fertilizers, 0.5=use genetically modified organisms (gmo)

"scale": 0.0=hunter-gatherers groups, 0.1=sparse villages/clans, 0.2=simple domain/tribes, 0.3=complex domain or archaic empire, 0.4=city-states or kingdom, 0.5=modern state or empire, 0.6=state-nation, 0.7=supranational union

"end": severity of polity collapse: 0=unknown, -0.1=slow decline (cultural/demographic or as result of external conquest), -0.2=natural disaster, -0.3=civil war or regime change

"en notes": textual summary of maximum 100 words about historical events happened in the polity. Focus on the following types of events: wars or battles, reforms, rulers, population, elites, disasters or epidemics, alliances or treaties, socio-conomic context, famines or financial stress, protests or movements, changes of elite, main type site, religions and philosophies

"references": report Google scholar references that support the information in previous columns. use grounding search to retrieve references to existing scientific papers. Here is an example of the desired output: { "pop": "0.7", "terr": "0.8", "war": "0.6", "rel": "0.5", "soc": "-0.1", "pow": "0.2", "infrastr": "0.5", "infomedia": "0.4", "economy": "0.3", "agri": "0.4", "scale": "0.5", "end": "0.1", "en notes": "Sultans Suleiman II and Ahmed II. Great Turkish War in the 1690s: the armies of the Holy League pushed the Ottomans back to the Balkans. end of expansionist policies in Europe, series of grand viziers from the Köprülü family (Köprülü era) were skilled administrators that revived the empire s fortunes after a period of military defeat and economic instability. Numerous reforms were instituted under their rule which enabled the empire to resolve its budget crisis., sultan Murad III (r.1574 - 1595). Morocco gained independence from Ottomans in 1582. Raising inflation and unrest between the Janissery (elite infantry). Start of Devsirme selection system to select bourocrats and military from a slave class educated to turkish culture. ,Sultan Selim II (r. 1566 - 1574 called Selim the drunk). Conquest of Cyprus in 1571 led to the naval defeat against the Holy League in the Battle of Lepanto", "references": "Wright, Elizabeth R. The Battle of Lepanto. Harvard Univ. Press, 2014." }

Figure 4: Prompt for the generation of synthetic historical data.

polity name required for the prompt and used as identifier in some seshat replication datasets such as the crisis consequence dataset.

The differences in the data generated by the three models is directly visible by plotting the TRVs in timeline charts. Figure 7 shows the average distribution of generated TRVs for each dimension and model (D is Deepseek, L is Llama and M is Mistral). Overall, the graphs reveal a consistent scoring across the TRVs from -11400 BCE to 2000 CE, and the models tend to converge mostly on the recent past. The LLMs used for the data generation demonstrate high inter-model correlation across some metrics – including war, economy, infrastructure, and population – but not others. Metrics like territory

```

"polityid": "17",
"polityname": "Hawaii I",
"start-year": "1000",
"end-year": "1200",
"politycode": "us-hawaii-1" "region": "Polynesia",
"regioncluster": "Polynesia",
"militlev": "0",
"adminlev": "0",
"statecode": "us",
"complexity": 0.203229,
"money": 1.8,
"infrast": 0.1,
"CapPop": 2.282858,
"setlev": "1",
"gov": 0.090909,
"PolTerr": 4.111759,
"PolPop": 3.854914,
"religlev": "0",
"texts": 0.1,
"agripod": 3.25,
"human-description": "Hawai'i, also known as the Big Island, is the largest island of the Hawaiian archipelago. Recent estimates for the date of initial settlement by Polynesian voyagers have varied from 800 to 1250 CE, but the latest Bayesian model, based on palaeoenvironmental data and a carefully defined set of archaeological radiocarbon dates, suggests that the archipelago was first colonized between 1000 and 1100... ",
"human-references": "Here are the references extracted from the text, formatted for clarity: - (Athens, Rieth and Dye 2014) J. Stephen Athens, Timothy M. Rieth and Thomas S. Dye. 2014. 'A Paleoenvironmental and Archaeological Model-Based Age Estimate for the Colonization of Hawai'i'. -American Antiquity- 79(1): 144-55. - (Kirch 2010, 127) Patrick V. Kirch. 2010. -How Chiefs Became Kings: Divine Kingship and the Rise of Archaic States in Ancient Hawai'i-. Berkeley, CA: University of California Press. - (Kirch 2012, 45) Patrick V. Kirch. 2012. -A Shark Going Inland Is My Chief-. Berkeley, CA: University of California Press. - ... ",

```

Figure 5: Example of data record in Seshat-eval (descriptions are reduced for the sake of space).

and particularly elite power type show the greatest divergence among the models, especially in the pre-Common Era, highlighting the fact that these dimensions are sensitive to cultural hacking.

The average text length of the generated descriptions is about 500 characters, with Llama 3.1 large having a higher standard deviation than the other LLMs. Llama is also the model that included in the description more explanations of the decisions about the scores reported in the TRVs. Seshat-eval includes different facets of the original Seshat Databank for comparison to the TRV-generated features. All numerical variables cover 27.9% of the polities present in Seshat-Perspective, except for the "crisis" variable, which covers 5.8% of the polities. The textual descriptions and references cover about 43.6% of the dataset.

4. Technical Validation

The quality of the generated data with respect to the human-coded data has been evaluated in three ways: measuring the correlation of numerical variables to the complexity score that is present in Seshat-eval; measuring the differences between the references generated by LLMs and selected by humans, and the similarity between textual descriptions.

Results, reported in Figure 8, show that most TRVs have similar correlation to complexity with respect to human-coded variables, except for government specialization (gov|pow), social structure (adminlev|soc) and polity collapse (crises|end). However, TRVs for agriculture and religion outperform agriculture productivity and religious levels in the correlation to complexity, indicating a better predictive power for cultural evolution analysis. Overall, without considering sign, the average correlation strength of human-coded variables is 0.7, while the average correlation strength of TRVs is 0.714.

The reference score has been validated with Jaccard similarity, comparing the human-coded references and the LLM-generated references. This method provides insights on the exact word match rate, and Llama 3.1 large shows the highest one. A manual evaluation on 100 examples annotated by assigning

```

"id": "17",
"name": "Hawaii1",
"new-name": "us-hawaii-1",
"long-name": "Hawaii I",
"start-year": "1000",
"end-year": "1200",
"duration": "200",
"avgcentury": 1100,
"regioncluster": "Polynesia",
"nga": "Big Island Hawaii",
"statecode": "us",
"capitalcity": "Kona",
"dr1pop": 0.3,
"dr1terr": 0.4,
"dr1war": 0.1,
"dr1rel": 0.2,
"dr1soc": -0.1,
"dr1pow": 0.1,
"dr1infrastr": 0.3,
"dr1infomedia": 0,
"dr1economy": 0.1,
"dr1agri": 0.3,
"dr1scale": 0.2,
"dr1end": 0,
"dr1notes": "Period of initial Polynesian settlement and population expansion. Society organized into dispersed, stratified family lineages ('ohana') led by chiefs ('ali i'). Economy based on sophisticated taro irrigation ('lo i kalo') and fishpond ('loko i a') aquaculture. Religion centered on ancestral and nature spirit worship ('aumakua'), with strict kapu (taboo) systems... ",
"dr1references": "Kirch, P. V. (2010). How Chiefs Became Kings: Divine Kingship and the Rise of Archaic States in Ancient Hawai i. University of California Press.; Hommon, R. J. (2013). The Ancient Hawaiian State: Origins of a Political Society. Oxford University Press.",
"m3mpop": 0.2,
"m3mterr": 0.4,
"m3mwar": 0.1,
"m3mrel": 0.4,
"m3msoc": -0.1,
"m3mpow": 0.1,
"m3minfrastr": 0.3,
"m3minfomedia": 0.1,
"m3meconomy": 0.1,
"m3magri": 0.2,
"m3mscale": 0.2,
"m3mend": 0,
"m3mnotes": "Between 1000 and 1200, Hawaiian society was organized into chiefdoms with stratified social structures. Chiefs (ali i) controlled land and resources, while commoners (maka āinana) engaged in agriculture and fishing. Warfare was limited to inter-chiefdom conflicts using stone tools and wooden weapons... ",
"m3mreferences": "Kirch, Patrick V. How Chiefs Became Kings: Divine Kingship and the Rise of Archaic States in Ancient Hawai i. University of California Press, 2010. Hommon, Robert J. The Ancient Hawaiian State: Origins of a Society. Oxford University Press, 2013.",
"l311pop": 0.2,
"l311terr": 0.3,
"l311war": 0.1,
"l311rel": 0.3,
"l311soc": -0.1,
"l311pow": 0.1,
"l311infrastr": 0.2,
"l311infomedia": 0.1,
"l311economy": 0.1,
"l311agri": 0.2,
"l311scale": 0.2,
"l311end": -0.1,
"l311notes": "Ancient Hawaiian society was divided into ali-i (chiefs) and commoners. The ali-i held power and controlled the land, while the commoners worked it. The islands were unified under a single ruler, likely around the 12th century. Agriculture was based on taro, sweet potatoes, and breadfruit, with fish and seafood supplementing the diet...",
"l311references": "Kirch, Patrick V. (2010). How Chiefs Became Kings: Divine Kingship and the Rise of Archaic States in Ancient Hawaii. University of California Press."

```

Figure 6: Example of generated data record (notes are reduced for the sake of space).

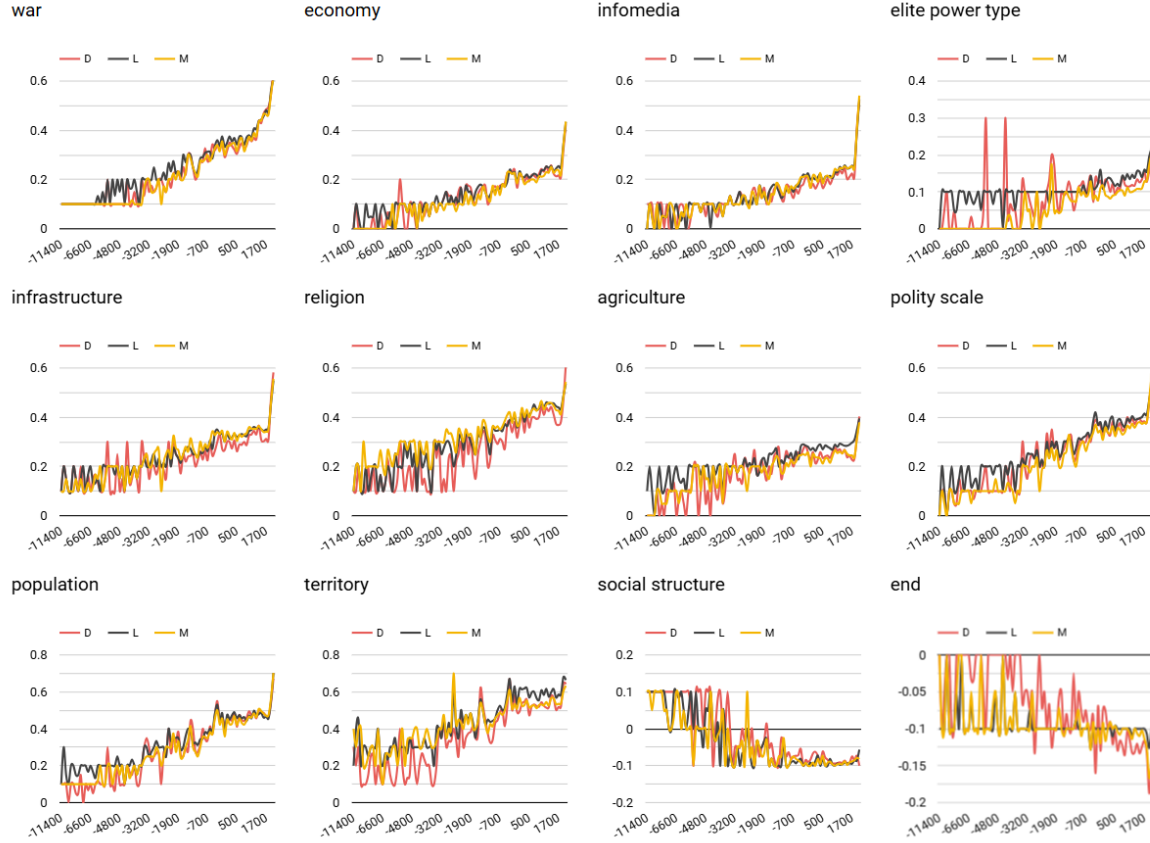


Figure 7: Distribution of the average TRV values generated by the three LLMs: Deepseek R1 (D), Llama 3.1-large (L) and Mistral 3 medium (M).

a score of 0 in case the generated references are not real or incomplete; 0.5 in case they are real but different from those selected by humans; 1 in the case at least one generated reference is the same selected by human experts and 0.7 in the case the generated references are different from the ones chosen by human experts, but all the LLMs generated the same reference. Results, show that all LLMs outperformed the baseline of 0.5 (the references exist but are different from those chosen by human experts). Overall, the models generated non-existent references in 5.6% of the cases.

The validation of the textual descriptions is based on the comparison between the generated text and random strings of 500 characters (baseline), and between the generated text and human-compiled text (result). The vocabulary is evaluated with Jaccard similarity, and the semantics is evaluated with Pairwise Cosine Similarity based on embeddings extracted with the all-MiniLM-L6-v2 transformer [24], that has 22.7M parameters and maps sentences and paragraphs to a 384 dimensional dense vector space. Results show that LLM performance is semantically homogeneous: among the three LLMs (Llama, Mistral, and Deepseek) there is little performance difference in cosine similarity gain over the random baseline in generating semantically similar descriptions, regardless of which embedding model is used. This suggests that the quality of the generated content, in terms of its ability to convey the correct historical information, is uniform across these three different generative models. Moreover, the evaluation with Jaccard similarity shows that Llama, having on average longer descriptions, has the highest rate of word matches with human compiled descriptions. In general, all LLMs proved to be efficient at generating polity descriptions that are similar, by vocabulary and semantics, to those written by human experts.

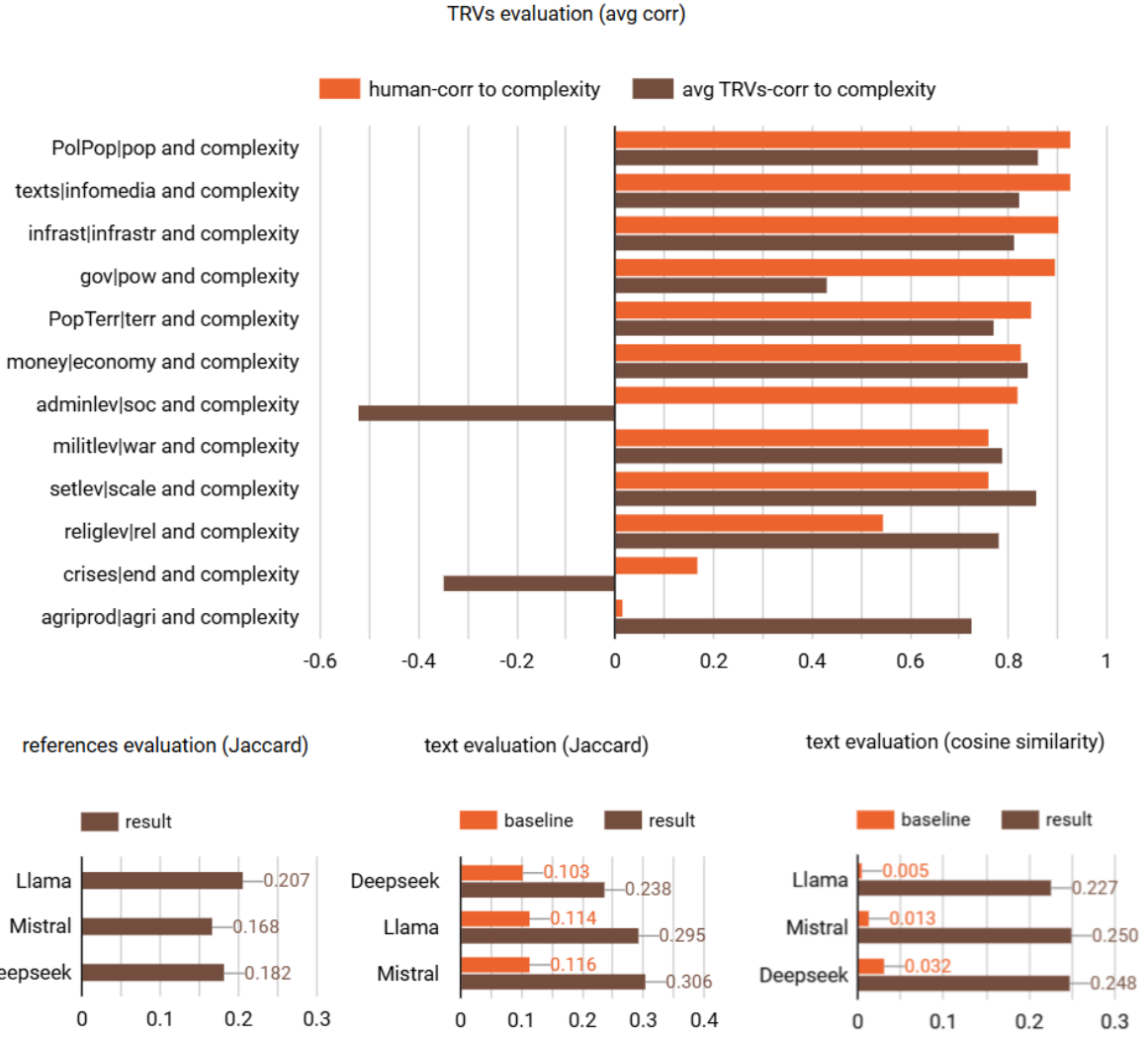


Figure 8: Results of the evaluation of Seshat-Perspective. TRVs are evaluated with correlation to complexity; references are evaluated with a human annotation; textual description evaluation compares the generated text against random noise (baseline) and human encoded descriptions (result) using pairwise cosine similarity and Jaccard similarity.

5. Conclusion

This paper introduced Seshat-Perspective, a novel historical databank annotated using a perspectivist approach with LLMs. By leveraging TRV prompting and deploying three prominent models from distinct geopolitical and cultural backgrounds, the original Seshat databank has been extended from 840 to 1023 polities. The technical validation demonstrated that synthetic data generation via LLMs is highly robust and semantically aligned with expert human annotation. Quantitative evaluation against the human-coded benchmark Seshat-eval yielded an average correlation strength to social complexity of 0.714 for the TRV dimensions, which marginally outperforms the human baseline of 0.70. Specifically, the TRV scales for agriculture and religion demonstrated superior correlation metrics to social complexity compared to their human-coded equivalents (agripod and religlev). Furthermore, semantic validation using pairwise cosine similarity with the all-MiniLM-L6-v2 transformer model indicated that the generated polity descriptions are semantically homogeneous and high-quality, yielding uniform textual descriptions across all three models that closely align with descriptions written by human experts. Reference tracking proved equally resilient; the models exhibited a remarkably low hallucination rate, with factual, verifiable references generated in 94.4% of evaluated cases. The use of

TRVs in a perspectivist framework provided evidence that, on average, Deepseek tends to underestimate population and territories, and interprets elites power types in a different manner than Llama and Mistral (see figure 7). This confirms previous findings [1] and the fact that cultural hacking is a subtle systematic bias. Incorporating comprehensive historical data into a perspectivist framework provides profound advantages for computational history and cultural analytics. For example, agreement and disagreement between models could be used as signals to identify cases for expert review. Moreover, a perspectivist historical databank offers ways to shift the conversation from searching for an impossible "objective" historical ledger toward quantifying, mapping, and managing subjectivity in expert data. Rather than enforcing a single top-down consensus, the perspectivist approach allows historians to test socio-cultural hypotheses against diverse synthetic viewpoints, or aggregate the quantitative responses to engineer culturally decentralized "gold standard" scores or labels.

To ensure absolute transparency and facilitate further exploration in cliodynamics, the resources developed in this work are fully open-source⁵. Ultimately, Seshat-Perspective serves as a scalable infrastructure for testing novel historical hypotheses with perspectivist approach. Researchers can seamlessly join external archaeological or geographical datasets by space-time metadata to calculate coefficients of determination against the multi-layered perspective tracks, or use the pipeline to benchmark entirely new time-resolved variables via automated prompting.

Declaration on Generative AI

During the preparation of this work, the authors used GPT-4 and Gemini 3 in order to perform: Grammar and spelling check; Paraphrase and reword; Improve writing style. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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This research employed data from the Seshat Databank (seshatdatabank.info) under Creative Commons Attribution Non-Commercial (CC By-NC SA) licensing.

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⁵https://colab.research.google.com/drive/1_4aUNGjl7_uhLZZKE7mAYHPhWYUZ9jvr?usp=sharing. The use of the notebook requires access to Huggingface APIs with a billing account.

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